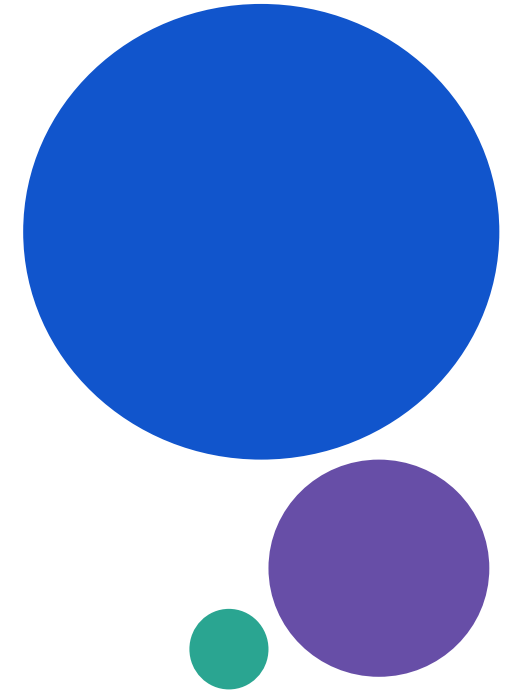

R-Powered Analytics in Power BI

From R to interactive business insight.



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(<https://ikhsannur1996.github.io/>)

R + Power BI

Two tools, complementary strengths

R



Powerful **open-source language for data analysis**, statistics, and advanced visualization.

Prepare • transform • model • visualize • forecast

POWER BI



Business intelligence platform for interactive dashboards, reporting, and data storytelling.

Interact • filter • communicate • distribute

R + Power BI |> Combine Power BI's usability with R's advanced analytics and custom visualizations.

Basic R

The essentials: objects, data types, vectors, functions & indexing

1 Objects & Values

```
x <- 10  
name <- "Ikhsan"  
score <- 87.5
```

Use `<-` to assign values. R objects can hold numbers, text, logical values, and more.

2 Data Types

```
num <- 10.5  
text <- "R"  
flag <- TRUE
```

Common basics: numeric, character, logical, and integer values.

3 Vectors

```
sales <- c(10,  
20, 30)  
sales * 2  
mean(sales)
```

Vectors store multiple values of the same basic type.

4 Functions

```
sum(sales)  
max(sales)  
length(sales)
```

Functions take inputs and return results. Use `()` to call a function.

5 Indexing

```
sales <- c(10, 20, 30, 40)  
sales[1]      # 10  
sales[2:3]    # 20, 30
```

R indexing starts at 1 — not 0.

Mini Example

```
sales <- c(120, 80, 150)  
mean(sales)  
sales[sales > 100]
```

Create → calculate → filter

R Libraries

Packages extend base R with ready-to-use functions and workflows.

tidyverse

Data manipulation & visualization

dplyr • ggplot2 • tidyr

dplyr

Data transformation

select • group • mutate

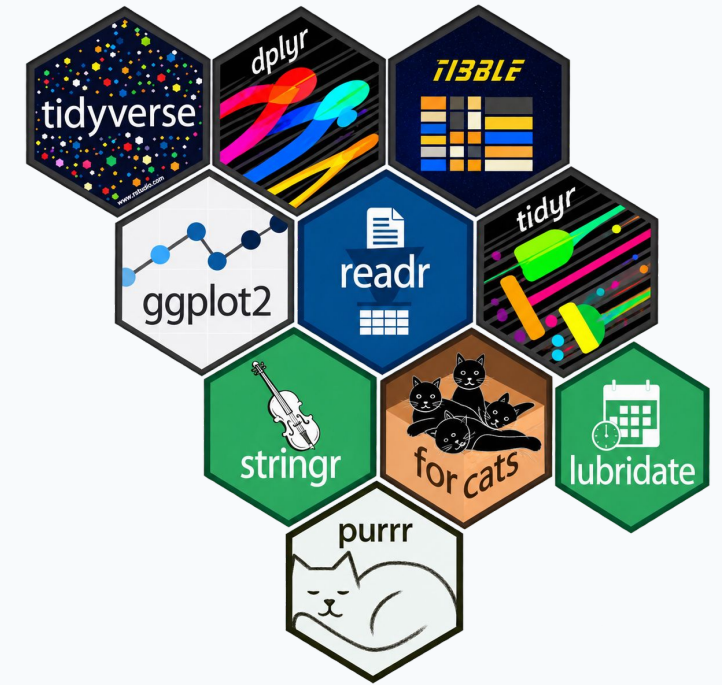
1 INSTALL

```
install.packages("tidyverse")
```

2 LOAD

```
library(tidyverse)
```

Install once • Load in each R session • Then use the package functions



R Pipe Operators

A pipe sends the result of one step directly into the next step.

%>%

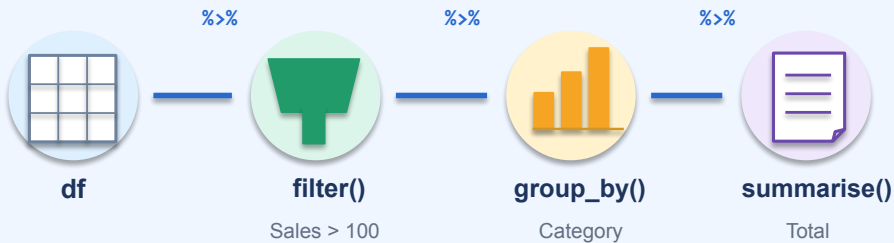
magrittr pipe

magrittr

Result → next function

```
library(dplyr)
```

```
df %>%  
  filter(Sales > 100) %>%  
  select(Product, Sales)
```



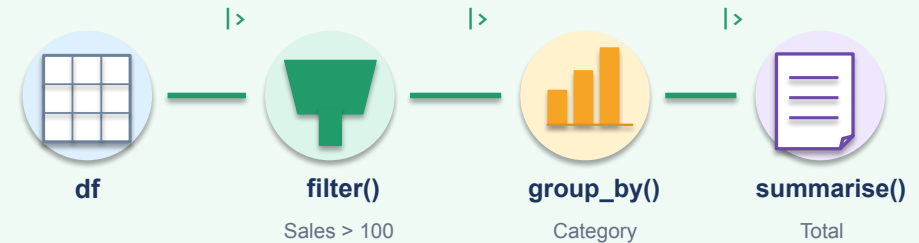
|>

native R pipe

R. 4.1.0

Result → next function

```
df |>  
  filter(Sales > 100) |>  
  select(Product, Sales)
```



Same concept: both pass the previous result forward — only the syntax and origin differ.

Basic dplyr

The core verbs for transforming and summarising data

Data manipulation

select()

```
df |>
  select(Name,
    Sales)
```

Choose columns.

filter()

```
df |>
  filter(Sales >
    100)
```

Choose rows.

mutate()

```
df |>
  mutate(Revenue =
    Price * Qty)
```

Create or modify columns.

arrange()

```
df |>
  arrange(desc(Sales))
```

Sort rows.

summarise()

```
df |>
  summarise(Avg =
    mean(Sales))
```

Reduce data to summaries.

group_by()

```
df |>
  group_by(Category) |>
  summarise(Total =
    sum(Sales))
```

Summarise by groups.

Mini Pipeline

```
sales |>
  filter(Sales > 100) |>
  mutate(Revenue = Price * Qty) |>
  group_by(Category) |>
  summarise(Total = sum(Revenue))
```

Filter → create → group → summarise

Tip: Learn the verbs first — then combine them with |> to build readable data workflows.

Basic ggplot2

Build visualisations by combining data, aesthetics, geoms and layers

Data → Visual

ggplot()

```
ggplot(df)
```

Start with a dataset.

aes()

```
aes(x = Month,  
    y = Sales)
```

Map variables to visuals.

geom_*()

```
+ geom_line()
```

Choose the chart type.

labs()

```
+ labs(title =  
       "Monthly  
       Sales")
```

Add titles and labels.

theme_*()

```
+  
theme_minimal()
```

Control the appearance.

The ggplot2 Recipe

```
ggplot(data, aes(x, y)) +  
  geom_point() +  
  labs(title = "Sales") +  
  theme_minimal()
```

Data → Mapping → Geometry → Labels → Theme

Common Geoms

geom_point()	→ scatter
geom_line()	→ trend
geom_bar()	→ categories
geom_col()	→ values

Change the geom to change the visual story.

R in Power BI

[R + Power BI](#)

Where R adds statistical power — and where Power BI native features are simpler

Typical flow

Power BI



R Script



Analysis



Visual

✓ Pros

- Advanced statistics & machine learning
- Flexible data transformation with R packages
- Custom visualisations with ggplot2
- Useful for specialised analytical workflows
- Can extend Power BI beyond built-in analytics

△ Cons

- Requires R installation and environment setup
- More technical than native Power BI features
- R visuals have interaction / formatting limits
- Refresh and gateway setup can be more complex
- Not every R workflow translates cleanly to Power BI

Use R when... the analysis or visualisation needs go beyond what native Power BI + DAX can comfortably provide.

Business Case

Demand & Capacity What-If Analysis

How does demand uncertainty affect production capacity and revenue?

THE SITUATION

Monthly demand changes with seasonality, product preference, location, and demand growth uncertainty.

THE CHALLENGE

Demand and capacity need to be analyzed together to reveal utilization, capacity gaps, and shortage risk.

THE DECISION

Production planning needs evidence for capacity, scheduling, mitigation, and revenue planning.

ANALYSIS OUTPUT

Demand • Capacity • Utilization • Capacity Gap • Revenue Impact • SKU Risk • Forecast

Follow This GitHub for Hands-on

Use the repository as the practical companion for the workshop.

GITHUB

github.com/ikhsannur1996/r-power-bi

HANDS-ON FLOW

01

R Basics

02

Transform

03

What-If

04

Visualize

05

Forecast

06

Power BI

Open the README → follow the project flow → copy the R code → run the analysis → build the Power BI output.

Dashboard Sample

Example of how the analysis can be operationalized in Power BI



Conclusion

What the workshop connects from code to business value

01

Start with the business question

Demand uncertainty, capacity constraints, and revenue impact define the analysis.

02

Build a reusable analytical flow

R handles preparation, transformation, scenarios, visualization, and forecasting.

03

Make scenarios interactive

Power BI slicers connect scenario selection directly to the R visual.

04

Turn analysis into action

Use capacity gaps, SKU risk, and forecast signals to support planning decisions.

The goal is not only to create a visual — it is to turn data into a decision.

THANK YOU

From Demand to Decisions

R-Powered Analytics in Power BI

Better code creates better analysis.
Better analysis creates better decisions.

R • Tidyverse • Power BI

[Hands-on: github.com/ikhsannur1996/r-power-bi](https://github.com/ikhsannur1996/r-power-bi)

